

PUMS ↔ AutoCount Integration — API Specification

Audience: PUM System Developer

This document defines the API contracts between PUMS and the AutoCount integration. There are **four endpoints across two directions**:

Name	Endpoint	Hosted by	Direction	Method	Machine Scope
API 1	/api/meter-reading/online	PUMS	AutoCount → PUMS	POST	ONLINE machines
API 2	/api/meter-reading/offline	PUMS	AutoCount → PUMS	POST	OFFLINE machines
Webhook 1 — Stock Issue Request	/api/stockissue	AutoCount webhook — https://atpwebhook.primatechnology.com.my	PUMS → AutoCount	POST	—
Webhook 2 — Stock Transfer Request	/api/stocktransfer	AutoCount webhook — https://atpwebhook.primatechnology.com.my	PUMS → AutoCount	POST	—

API 1 and **API 2** are implemented **by PUMS** (AutoCount calls them). **Webhook 1** and **Webhook 2** are consumed **by PUMS** (PUMS calls the AutoCount webhook).

API 1 — Meter Reading (Machine Status: ONLINE)

Scope: Only machines with `Machine Status = ONLINE` are in scope for this API. Reference: http://atgroup.asia/all_device/ — filter the device list to ONLINE machines only. Any code in the request whose machine is not ONLINE must be omitted from the response.

Implemented by: PUMS

Method: POST

Path: /api/meter-reading/online

Content-Type: application/json

Returns the **latest** BK (Black) and CL (Color) meter readings for a list of ONLINE machines. Tracking ID is **not** returned by this API — it is handled by API 2.

POST is used (not GET) because the list of codes can be large — up to thousands of machines per billing run. Sending them in a JSON body avoids URL length limits and keeps the contract simple (one call, no batching, no URL-encoding of spaces).

Request — JSON Body

Field	Type	Required	Sample Value	Description
codes	string[]	Yes	["CSSI00002153", "CSSI00000090", "ASSI00000070"]	List of machine codes to query. May contain up to thousands of items.

```
{
  "codes": ["CSSI 00002153", "CSSI 00000090", "ASSI 00000070"]
}
```

Response — 200 OK

JSON array. One object per ONLINE machine code requested. Codes whose machine is not ONLINE are omitted.

For each machine, return the **latest** meter reading available (most recent audit), regardless of date.

Field	Type	Sample Value	Description
Code	string	CSSI 00002153	Machine code (echo of the requested code).
SerialNumber	string	XZL02175	Machine serial number.
TotalBK	integer	17257	Latest Total Black meter reading.
TotalCL	integer	11978	Latest Total Color meter reading.
LastAuditDate	string (ISO-8601)	2026-05-12T15:01:32	Date/time of the latest audit reading.

Example

Request

```
POST /api/meter-reading/online
Content-Type: application/json

{
  "codes": ["CSSI 00002153", "CSSI 00000090", "ASSI 00000070"]
}
```

Response

```
[
  {
    "Code": "CSSI 00002153",
    "SerialNumber": "XZL02175",
    "TotalBK": 17257,
    "TotalCL": 11978,
    "LastAuditDate": "2026-05-12T15:01:32"
  },
  {
    "Code": "CSSI 00000090",
    "SerialNumber": "JBA05089",
    "TotalBK": 1562075,

```

```
"TotalCL": 41833,  
  "LastAuditDate": "2026-05-13T14:58:05"  
}  
]
```

API 2 — Meter Reading (Machine Status: OFFLINE)

Scope: Only machines with `Machine Status = OFFLINE` are in scope for this API. Reference: http://atgroup.asia/tasks_meter/ — the readings come from the **Tasks Meter** report, which is how OFFLINE machines have their meters captured. Any code in the request whose machine is not OFFLINE must be omitted from the response.

Implemented by: PUMS

Method: POST

Path: `/api/meter-reading/offline`

Content-Type: `application/json`

Returns the **latest** BK (Black) and CL (Color) meter readings **within the requested month** for a list of OFFLINE machines, plus the **required** Task Tracking ID for that machine in that month.

POST is used (not GET) because the list of codes can be large — up to thousands of machines per billing run. Sending them in a JSON body avoids URL length limits and keeps the contract simple (one call, no batching, no URL-encoding of spaces).

Request — JSON Body

Field	Type	Required	Sample Value	Description
codes	string[]	Yes	<code>["CSSI 00002153", "CSSI 00000090", "ASSI 00000070"]</code>	List of machine codes to query. May contain up to thousands of items.
month	int (1-12)	Yes	<code>3</code>	Billing month.

```
{  
  "codes": ["CSSI 00002153", "CSSI 00000090", "ASSI 00000070"],  
  "month": 3  
}
```

Response — 200 OK

JSON array. One object per OFFLINE machine code requested. Codes whose machine is not OFFLINE — **or** for which no Tracking ID can be resolved for the requested month — are omitted from the response (do **not** return them with `TrackingId = null`).

For each machine, return the **latest** meter reading **within the requested month** (most recent audit in that month).

Field	Type	Sample Value	Description
Code	string	CSSI 00002153	Machine code (echo of the requested code).
SerialNumber	string	XZL02175	Machine serial number.
TotalBK	integer	17257	Latest Total Black meter reading in the requested month.
TotalCL	integer	11978	Latest Total Color meter reading in the requested month.
LastAuditDate	string (ISO-8601)	2026-05-12T15:01:32	Date/time of the latest audit reading in the requested month.
TrackingId	string (required, non-null, prefix MR)	MR-260512-008	Latest Task Tracking ID for this machine in the requested month. Always present in API 2, and must start with MR .

Example

Request

```
POST /api/meter-reading/offline
Content-Type: application/json

{
  "codes": ["CSSI 00002153", "CSSI 00000090", "ASSI 00000070"],
  "month": 5
}
```

Response

```
[
  {
    "Code": "CSSI 00002153",
    "SerialNumber": "XZL02175",
    "TotalBK": 17257,
    "TotalCL": 11978,
    "LastAuditDate": "2026-05-12T15:01:32",
    "TrackingId": "MR-2026-0512-008"
  },
  {
    "Code": "CSSI 00000090",
    "SerialNumber": "JBA05089",
    "TotalBK": 1562075,
    "TotalCL": 41833,
    "LastAuditDate": "2026-05-13T14:58:05",
    "TrackingId": "MR-2026-0513-014"
  }
]
```

Webhook 1 — Stock Issue Request (PUMS → AutoCount)

Called by: PUMS

Method: POST

URL: https://atpwebhook.prismatechnology.com.my/api/stockissue

Content-Type: application/json

Reference: <https://atgroup.asia/stockReport/> — source of the Stock Issue records on the PUMS side.

PUMS pushes **one** Stock Issue record to AutoCount per call.

Request Body — JSON object

Field	Type	Description
StockIssueId	string	Unique ID generated by PUMS. Used for idempotency.
IssueDateTime	string (ISO-8601)	Issue date/time.
StockIssueNo	string	Stock Issue document number.
ReferenceNo	string	Reference number.
Description	string	Description.
Department	string	Department code.
Job	string	Job code.
Technician	string	Technician code / name.
Location	string	Stock location code.
ItemCode	string	Item code.
Quantity	number	Quantity issued.
UOM	string	Unit of measure.

```
{
  "StockIssueId": "PUMS-SI-000123",
  "IssueDateTime": "2026-01-01T00:00:00",
  "StockIssueNo": "DOS2601/001",
  "ReferenceNo": "RPS-251230-043",
  "Description": "Stock Issue-REDHA",
  "Department": "A/4PE20147",
  "Job": "MC001",
  "Technician": "AUTOCOUNT",
  "Location": "PERLING",
  "ItemCode": "NPG89 T/BK",
  "Quantity": 1,
  "UOM": "PCS"
```

```
}
```

Response

Success — 200 OK

```
{
  "success": true,
  "stockIssueId": "PUMS-SI-000123",
  "message": "Stock Issue task received."
}
```

Failure — non-2xx (e.g. 400, 409, 422, 500)

```
{
  "success": false,
  "stockIssueId": "PUMS-SI-000123",
  "errorCode": "INVALID_PAYLOAD",
  "reason": "Field 'Quantity' is required."
}
```

Field	Type	Description
success	boolean	true on accept, false on reject.
stockIssueId	string	Echo of the request stockIssueId.
message	string	Human-readable success note (present only when success = true).
errorCode	string	Machine-readable error code (see table below). Present only when success = false.
reason	string	Human-readable explanation of the failure. Present only when success = false.

Error Codes

HTTP Status	errorCode	Meaning
400	INVALID_PAYLOAD	Required field missing or wrong type.
500	INTERNAL_ERROR	Unexpected server error. Retry recommended.

Duplicates are allowed. Re-sending the same stockIssueId is **not** an error — the previously received payload for that ID is overwritten with the new one. PUMS may safely re-submit without checking whether a stockIssueId has been sent before. (Note: this endpoint only **receives** the data; it does not auto-create an AutoCount Stock Issue document. Document creation is handled separately on the AutoCount side.)

Webhook 2 — Stock Transfer Request (PUMS → AutoCount)

Called by: PUMS

Method: POST

URL: https://atpwebhook.prismatechnology.com.my/api/stocktransfer

Content-Type: application/json

Reference: https://atgroup.asia/stockStandby/ — source of the Stock Transfer records on the PUMS side.

PUMS pushes **one** Stock Transfer record to AutoCount per call.

Request Body — JSON object

Field	Type	Description
RequestId	string	Unique ID generated by PUMS. Used for idempotency.
DocumentDateTime	string (ISO-8601)	Document date/time.
Technician	string	Technician code.
Part	string	Part / item code.
qty	number	Quantity.
type	string	Transfer type.
unit	string	Unit of measure.
approval	string	Approval reference.

```
{
  "RequestId":      "50460",
  "DocumentDateTime": "2026-05-06T11:18:09",
  "Technician":     "AUTOCOUNT",
  "Part":           "TC001 [USED PART]",
  "qty":            -2,
  "type":           "OUT",
  "unit":           "UNIT",
  "approval":       "Yes"
}
```

Response

Success — 200 OK

```
{
  "success": true,
  "requestId": "50460",
  "message": "Stock Transfer task received."
}
```

Failure — non-2xx (e.g. 400, 500)

```
{
  "success": false,
  "requestId": "50460",
  "errorCode": "INVALID_PAYLOAD",
  "reason": "Field 'qty' is required."
}
```

Field	Type	Description
success	boolean	true on accept, false on reject.
requestId	string	Echo of the request <code>RequestId</code> .
message	string	Human-readable success note (present only when <code>success = true</code>).
errorCode	string	Machine-readable error code (see table below). Present only when <code>success = false</code> .
reason	string	Human-readable explanation of the failure. Present only when <code>success = false</code> .

Error Codes

HTTP Status	errorCode	Meaning
400	INVALID_PAYLOAD	Required field missing or wrong type.
500	INTERNAL_ERROR	Unexpected server error. Retry recommended.

Master data and field values are not validated. Unknown `Part`, `Tech`, or `type` values are accepted as-is — AutoCount will not reject the document on those grounds. Make sure PUMS sends correct values; AutoCount won't tell you if they don't match.

Duplicates are allowed. Re-sending the same `RequestId` is **not** an error — the previously received payload for that ID is overwritten with the new one. PUMS may safely re-submit without checking whether a `RequestId` has been sent before. (Note: this endpoint only **receives** the data; it does not auto-create an AutoCount Stock Transfer document. Document creation is handled separately on the AutoCount side.)

Conventions

- **Encoding:** `UTF-8`. All bodies are JSON.

- **Timestamps:** ISO-8601, preferably UTC (`2026-03-12T08:45:00Z`).
- **One record per call:** Webhook 1 and Webhook 2 each accept a **single** JSON object per request (not an array). PUMS sends one HTTP call per Stock Issue / Stock Transfer.
- **Upsert semantics (duplicates allowed):** `StockIssueId` (Webhook 1) and `RequestId` (Webhook 2) act as the key for the received payload. Re-sending the same ID overwrites the previously received payload — it does **not** auto-create or modify any AutoCount document. Document creation is handled separately on the AutoCount side. PUMS does not need to track which IDs have been sent.
- **Response shape:**
 - On success → HTTP `2xx` with `{ "success": true, ... }`.
 - On failure → HTTP non-2xx with `{ "success": false, "errorCode": "...", "reason": "..." }`.
- **Retries:**
 - HTTP `5xx` (e.g. `INTERNAL_ERROR`) or transport failure → retry with exponential backoff (safe — re-send is an upsert).
 - HTTP `4xx` (e.g. `INVALID_PAYLOAD`) → do **not** retry blindly; the payload must be corrected first.
- **Authentication: API key** — all four endpoints (API 1, API 2, Webhook 1, Webhook 2) require an API key sent in the HTTP header `X-API-Key` . Keys are issued per integration partner and must be kept secret. Requests without a valid key return `401 Unauthorized` .

`X-API-Key: <your-api-key>`

Open Items

- API key exchange — PUMS and AutoCount sides each issue a key for the other to use.
- Field length / numeric precision limits.
- Base URL for API 1 and API 2 (both PUMS-hosted). The relative paths are defined in this document; only the host needs to be supplied.
- Final list of `errorCode` values (the tables above are a starting set).